A4.2 AXI3 memory attribute signaling

In AXI3, the **AxCACHE[3:0]** signals specify the *Bufferable*, *Cacheable*, and *Allocate* attributes of the transaction.

Table A4-1 shows the **AxCACHE** encoding.

Table A4-1 Transaction attribute encoding

AxCACHE	Value	Transaction attribute
[0]	0	Non-bufferable
	1	Bufferable
[1]	0	Non-cacheable
	1	Cacheable
[2]	0	No Read-allocate
	1	Read-allocate
[3]	0	No Write-allocate
	1	Write-allocate

AxCACHE[0], Bufferable (B) bit

When this bit is asserted, the interconnect, or any component, can delay the transaction reaching its final destination for any number of cycles.

Note

Normally, the Bufferable attribute is only relevant to writes.

AxCACHE[1], Cacheable (C) bit

When this bit is deasserted, allocation of the transaction is forbidden.

When this bit is asserted:

- Allocation of the transaction is permitted. RA and WA give additional hint information.
- The characteristics of a transaction at the final destination does not have to match the characteristics of the original transaction.

  For writes this means that a number of different writes can be merged together.

  For reads this means that the contents of a location can be prefetched, or the values from a single fetch can be used for multiple read transactions.

AxCACHE[2], Read-allocate (RA) bit

When this bit is asserted, read allocation of the transaction is recommended but is not mandatory.

The RA bit must not be asserted if the C bit is deasserted.

AxCACHE[3], Write-allocate (WA) bit

When this bit is asserted, write allocation of the transaction is recommended but is not mandatory.

The WA bit must not be asserted if the C bit is deasserted.

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